Networking Task 0.1:Understanding Network Environment

Part A: Network Information

Finding Network information using the command prompt in windows:
Type command ipconfig /all in the cmd and all information will be displayed on the screen

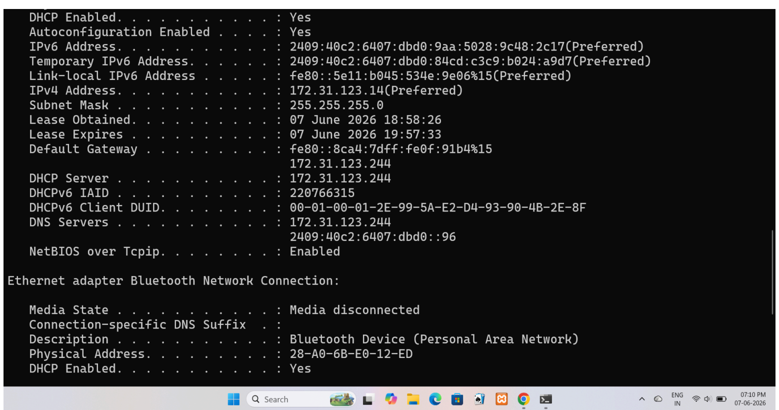
1.Hostname:
Device name is:Nilesh_Raverkar

2.Ipv4 address:
192.6.43.3

3.MAC address:
00-1A-2B-3C-4D-5F

4.DNS Server:
172.31.123.241

4. Default Gateway:
fe80::8ca4:7dff:fe0f:91b4%15



PART B :-Basic Networking concept:

1. IP Address (Internet Protocol Address)

An **IP Address** is a logical address assigned to a device on a network. It uniquely identifies a device and allows data to be sent and received between devices.

Why is an IP Address Needed?

When you open a website, your device sends a request to a web server. The server needs to know:

- Who sent the request?
- Where should the response be sent?

The IP address provides this information.

2. MAC Address (Media Access Control Address)

A **MAC Address** is a unique physical address assigned to a network interface card (NIC) by the manufacturer.

Example:

00:1A:2B:3C:4D:5E

- First 24 bits → Manufacturer Identifier
- Last 24 bits → Device Identifier

3. Default Gateway

A **Default Gateway** is the device that forwards traffic from one network to another.

Why is it needed?

Suppose:

- Your laptop: 192.168.1.10
- Router: 192.168.1.1
- Website server: 142.250.193.78

Your laptop knows:

- The website is not inside the local network.
- It needs help reaching another network.

So it sends the data to the default gateway (router).
The router then forwards the traffic to the Internet.

4. DNS (Domain Name System)

DNS is a system that converts domain names into IP addresses.

Humans remember names easily, but computers communicate using IP addresses.

Example

You type:

www.google.com

DNS converts it into:

142.250.x.x

5. Difference Between Public IP and Private IP

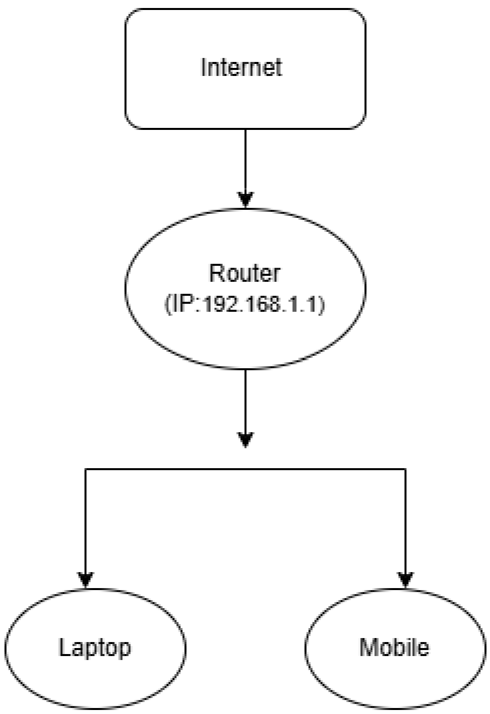
Public IP Address

- A Public IP is an IP address that is visible on the Internet.
- It is assigned by your **Internet Service Provider (ISP)** and is unique across the entire Internet.
- Characteristics of Public IP :Globally unique,Assigned by ISP ,Accessible from the Internet .

Private IP Address

- A **Private IP Address** is used only within a local network (LAN).
- These addresses are not directly accessible from the Internet.
- Characteristics :Used inside local networks,Not visible on the Internet.

PART C:Basic Network Diagram:



Part D:Network connectivity test:

Using Windows:

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PS C:\Users\rmrav> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::e2b2:3cd6:8dee:d2e9%16
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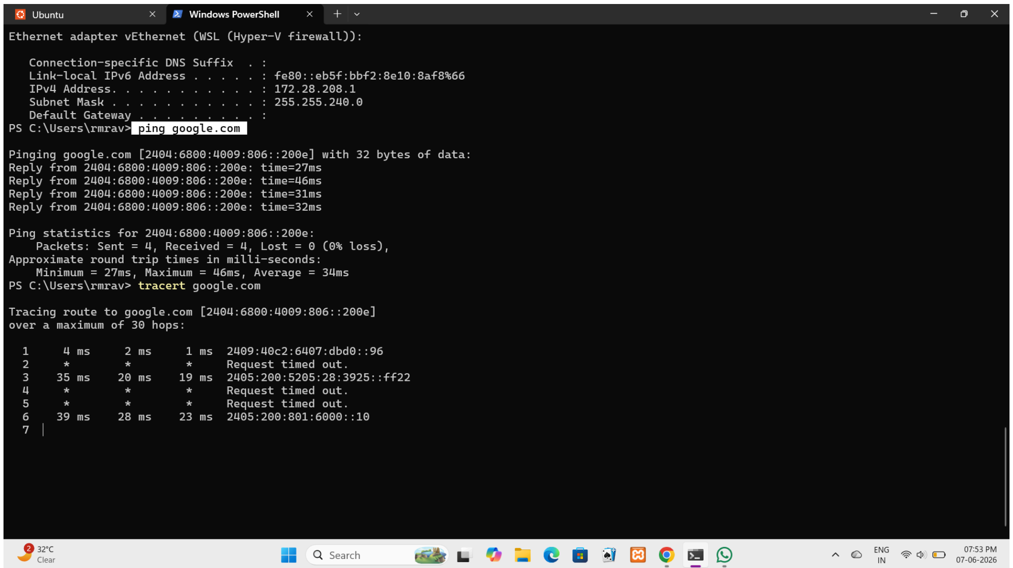

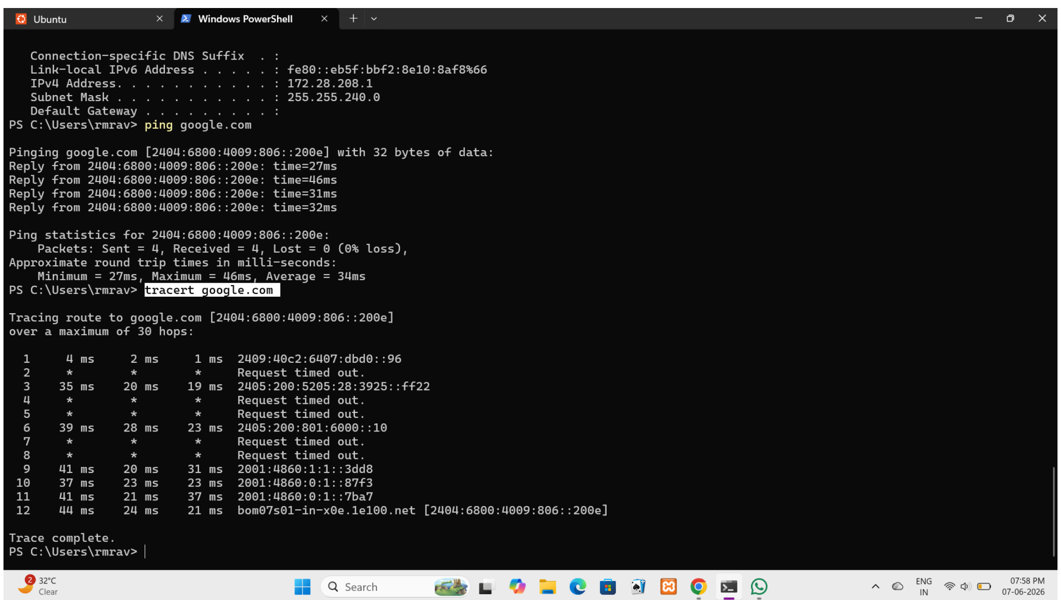

Questions”

1.Was ping successful?

Ans:-Yes

2.How many hops were shown?

Ans:12 hops.

3.What is the purpose of the traceroute?

Ans:Traceroute is used to trace the path taken by data packets from a source device to a destination and to identify network connectivity or routing problems.